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Inventor(s)	Choi; Hyungho et al.

Chemical supply system

Abstract

A chemical supply system, includes: a chemical container receiving station configured to receive a chemical container, inspect chemical information of the chemical container and confirm the chemical information; a temporary storage configured to store the chemical container; a chemical supply device configured to supply a chemical in the chemical container to a place of use of the chemical; a first robot configured to transport the chemical container for which the chemical information is confirmed from the chemical container receiving station; and a second robot configured to carry the chemical container transported by the first robot into the chemical supply device.

Inventors: Choi; Hyungho (Suwon-si, KR), Kim; Dongwon (Suwon-si, KR), Kim; Minyoung (Hwaseong-si, KR), Kim; Jongkyu (Suwon-si, KR), So; Jinho (Seoul, KR), Yang; Jaehwa (Hwaseong-si, KR), Hwang; Soonwook (Suwon-si, KR)

Applicant: Samsung Electronics Co., Ltd. (Suwon-si, KR)

Family ID: 1000008821690

Assignee: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

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Primary Examiner: Myers; Glenn F

Attorney, Agent or Firm: Muir Patent Law, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Korean Patent Application No. 10-2022-0027536, filed on Mar. 3, 2022, with the Korean Intellectual Property Office, and 10-2022-0047356, filed on Apr. 18, 2022, with the Korean Intellectual Property Office, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

1. Field

(2) The present disclosure relates to a chemical supply system.

2. Description of Related Art

(3) Various types of chemicals used in a semiconductor process are respectively supplied to appropriate locations (or process operations) by a chemical supply device. The chemicals are supplied in a chemical container, and therefore, an operation of carrying in a chemical container into the chemical supply device or carrying out a chemical container from which chemicals are exhausted from the chemical supply device is performed. In the carrying-in and carrying-out operations of a chemical container, it is important to develop a chemical supply system that can prevent safety accidents and improve operational efficiency.

SUMMARY

(4) An aspect of the present inventive concept is to provide a chemical supply system having improved operational efficiency.

(5) According to an aspect of the present inventive concept, a chemical supply system, includes: a chemical container receiving station configured to receive a chemical container, inspect chemical information of the chemical container and confirm the chemical information; a temporary storage configured to store the chemical container; a chemical supply device configured to supply a chemical in the chemical container to a place of use of the chemical; a first robot configured to transport the chemical container for which the chemical information is confirmed from the chemical container receiving station; and a second robot configured to carry the chemical container transported by the first robot into the chemical supply device.

(6) According to an aspect of the present inventive concept, a chemical supply system, includes: a chemical supply device having a chemical container accommodating space in which a chemical container is accommodated, and configured to supply a chemical in the chemical container to a place of use of the chemical; and a mobile robot configured to fasten and separate a coupling interface to an outlet of the chemical container carried into the chemical accommodating space.

(7) According to an aspect of the present inventive concept, a chemical supply system, includes: a chemical container receiving station configured to inspect chemical information of a chemical container and confirm the chemical information; a temporary storage configured to store the chemical container; a chemical supply device configured to supply a chemical in the chemical container to a place of use of the chemical; first and second robots configured to transfer the

chemical container for which the chemical information is confirmed to the temporary storage, or the chemical supply device, wherein the first robot is configured to transfer the chemical container from the chemical container receiving station to the temporary storage or transfer the chemical container to the second robot, and the second robot is configured to transfer the chemical container received from the first robot to the chemical supply device.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The above and other aspects, features, and advantages of the present inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a diagram illustrating a chemical supply system according to an example embodiment of the present inventive concept.
- (3) FIGS. 2A and 2B are diagrams exemplarily illustrating a chemical carrying-in device applicable to a chemical supply system according to an example embodiment.
- (4) FIGS. 3A and 3B are diagrams exemplarily illustrating a first robot applicable to a chemical supply system according to an example embodiment.
- (5) FIGS. 4A and 4B are diagrams exemplarily illustrating a temporary storage applicable to a chemical supply system according to an example embodiment.
- (6) FIGS. 5A and 5B are diagrams exemplarily illustrating a second robot applicable to a chemical supply system according to an example embodiment.
- (7) FIGS. 6A and 6B are diagrams exemplarily illustrating a chemical supply device applicable to a chemical supply system according to an example embodiment.
- (8) FIGS. 7A to 7D are diagrams exemplarily illustrating a third robot applicable to a chemical supply system according to an example embodiment.
- (9) FIG. 8 is a diagram exemplarily illustrating a chemical supply system comprising the exemplary devices shown in FIGS. 2A to 6B.

DETAILED DESCRIPTION

- (10) Hereinafter, example embodiments of the present inventive concept will be described with reference to the accompanying drawings as follows.
- (11) FIG. 1 is a diagram illustrating a chemical supply system **100** according to an example embodiment of the present inventive concept.
- (12) Referring to FIG. 1, the chemical supply system **100** of an example embodiment may include a chemical carrying-in device **110**, a temporary storage **120**, a chemical supply device **130**, a first robot **140**, and a second robot **150**. In chemical supply system **100** according to the present inventive concept, by putting a chemical container bundle **10** carried into the chemical carrying-in device **110** in the chemical supply device **130** using the first robot **140** and the second robot **150**, or storing the same in the temporary storage **120**, it is possible to prevent a safety accident occurring in a process of carrying in and carrying out a heavy chemical container, and to improve efficiency of a transfer operation of the chemical container. It should be noted that certain items are described in the singular, but are provided in plural, as can be seen from the various figures. Also, ordinal numbers such as “first,” “second,” “third,” etc. may be used simply as labels of certain elements, steps, etc., to distinguish such elements, steps, etc. from one another. Terms that are not described using “first,” “second,” etc., in the specification, may still be referred to as “first” or “second” in a claim. In addition, a term that is referenced with a particular ordinal number (e.g., “first” in a particular claim) may be described elsewhere with a different ordinal number (e.g., “second” in the specification or another claim).
- (13) The chemical carrying-in device **110** may be configured to inspect chemical information

included in the chemical container bundle **10**. The chemical information may include information for determining whether a chemical in the chemical container is a chemical used by the chemical supply device **130** or a place of use thereof, and may include, for example, material information, an expiration date, and the like. The chemical information may be inspected by scanning a code, such as a barcode or a QR code attached to a surface of the chemical container. For example, when the chemical information of the chemical container states that the chemical thereof is unsuitable for use in the chemical supply device **130**, the chemical container may be returned externally.

(14) The temporary storage **120** may be configured to store a chemical container. The temporary storage may temporarily store a chemical bundle **10** before being carried into the chemical supply device **130** and/or a chemical bundle **10** being carried out from the chemical supply device **130** after all the chemicals thereof are exhausted. The temporary storage **120** may be disposed between the chemical carrying-in device **110** and the chemical supply device **130** in consideration of movement of the first robot **140** and the second robot **150**. For example, the chemical carrying-in device **110** may be disposed on one side (e.g., a first side) of the temporary storage **120**, and the chemical supply device **130** may be disposed on the other side (e.g., a second side) of the temporary storage **120**.

(15) The chemical supply device **130** may be configured to supply the chemical in the chemical container included in the chemical container bundle **10** to a place of use thereof. The chemical supply device **130** may connect a coupler to an outlet of the chemical container, so that the chemical in the chemical container is supplied to the place of use thereof along a chemical supply pipe. For example, a pump for chemical transfer, or the like may be installed in the chemical supply device **130**. The chemical supply device **130** may be described as a chemical supplying device, or a chemical supplying station.

(16) The first robot **140** and the second robot **150** may be configured, programmed, and controlled to transfer a chemical container bundle **10** or a chemical container for which the chemical information is confirmed to the temporary storage **120** or the chemical supply device **130**. The first robot **140** may be configured, programmed, and controlled to carry out the chemical container bundle **10** or the chemical container from the chemical carrying-in device **110**. The second robot **150** may be configured, programmed, and controlled to carry in the chemical container bundle **10** or the chemical container carried out by the first robot into the chemical supply device **130**. For example, the first robot **140** may transfer the chemical container bundle **10** or the chemical container from the chemical carrying-in device **110** to the temporary storage **120** or transfer the same to the second robot **150**, and the second robot **150** may transfer the chemical container bundle **10** or the chemical container received from the first robot **140** to the chemical supply device **130**. The first robot **140** and the second robot **150** may perform an operation according to a transfer command or replacement command of the chemical container transmitted by a control system. The first robot **140** and the second robot **150** may access the chemical carrying-in device **110**, the temporary storage **120**, or the chemical supply device **130** designated by the control system, and may perform a transfer operation of the chemical container bundle **10**. The first robot **140** and the second robot **150** may perform a transfer operation of the chemical container bundle **10** according to location information and/or path information input in advance. As described above, the first robot **140** and the second robot **150** interact to transfer the chemical container bundle **10** or chemical container, thereby preventing safety accidents and improving the efficiency of the transfer operation of the chemical container. The first robot **140** may be a first type of robot having a first structure and configuration, and the second robot **150** may be a second type of robot different from the first robot in structure and/or configuration.

(17) Hereinafter, each component that may be applied to the chemical supply system **100** according to example embodiments of the present inventive concept will be described in detail.

(18) FIGS. 2A and 2B are diagrams exemplarily illustrating a chemical carrying-in device **110** applicable to a chemical supply system according to an example embodiment.

(19) Referring to FIGS. 2A and 2B, the chemical carrying-in device **110** of an example embodiment may include a transfer member **111** and an inspection member **112**. The chemical carrying-in device **110** is a location in which the chemical container **11** or the chemical container bundle **10** containing the chemical is received, and may be described as a chemical carrying-in station or a chemical container receiving station. The chemical container bundle **10** may be put into the chemical carrying-in device **110** by a forklift or an unmanned forklift. The chemical container bundle **10** may include a plurality of chemical containers **11**, for example drums, loaded on the pallet **12**. Hereinafter, the term “chemical container **11**” may be used to describe a single, individual chemical container (e.g., drum), a plurality of chemical containers, or a chemical container bundle **10** loaded on a pallet **12**. The chemical carrying-in device **110** (e.g., the transfer member **111** of the chemical carrying-in device **110**) may transfer the chemical container **11** received from the outside to the inspection member **112**, and chemical information of the chemical container **11**, for example, material information, an expiration date, and the like may be inspected with the inspection member **112**. The chemical container **11** for which the chemical information is confirmed may be carried out (e.g., removed and transported) from the chemical carrying-in device **110** by the first robot **140** (refer to FIG. 3A).

(20) The transfer member **111** may have one side (e.g., a first side) on which the chemical container **11** is seated and the other side (e.g., a second side) on which the chemical container **11** is transferred to inspect chemical information. The transfer member **111** may be configured to transfer the chemical container bundle **10** loaded on one side in one direction. For example, when the pallet **12** on which the chemical container **11** is loaded is seated on the one side of the transfer member **111**, the transfer member **111** may be operated so that the pallet **12** is transferred to the other side, and when the pallet **12** is seated on the other side of the transfer member **111**, the transfer member **111** may be stopped.

(21) The transfer member **111** may be a transfer platform configured to transport a chemical container **111** from one location to a second location. For example, in one embodiment, the transfer member **111** may be a conveyor device, composed of rollers or belts **111a**. For example, the transfer member **111** may transport the chemical container **11** from the one side to the other side or an opposite direction thereof by rotating a plurality of rollers disposed in one direction to move a belt on which the chemical container **11** is mounted. However, other transport mechanisms may be used, such as hydraulic or actuator controlled movable arms or frames.

(22) According to an example embodiment, a first sensor (not illustrated) for detecting whether the chemical container **11** or the pallet **12** is seated on the transfer member **111** may be disposed. As an example, in order to check whether the chemical container **11** is seated and a position of the transfer member **111**, the first sensor may be installed at regular intervals on a side surface of the transfer member **111** or on a fall prevention guide. The first sensor may be, for example, a load detection sensor for detecting the presence or absence of the chemical container **11** on the transfer member **111** by detecting an amount of load (e.g., weight) disposed on the transfer member **111**). However, other sensors may be used, such as image sensors, for example. In addition, the transfer member **111** may be made of a corrosion-resistant coating, which is strongly resistant to corrosion, and a corrosion-resistant material such as PTFE, or the like, and according to an example embodiment, a leakage detection sensor may be installed on the transfer member **111** to detect chemical leakage of the chemical container **11**. For example, a leakage detection sensor may sense the presence of liquid.

(23) Meanwhile, the chemical carrying-in device **110** may further include an alignment member **113** configured to align the chemical container bundle **10** on the transfer member **111**. The alignment member **113** may be disposed on both sides (e.g., opposite sides) of the transfer member **111** and in a direction of movement of the pallet **12**, to align the chemical container **11** or the pallet **12** to a central portion of the transfer member **111**. For example, the alignment member **113** may align the chemical container **11** to the central portion of the transfer member **111** before the transfer

member **111** operates. In addition, a guide for preventing separation of the chemical container **11** or the chemical container bundle **10** may be installed on both sides of the transfer member **111** while the transfer member **111** operates.

(24) Depending on a transfer distance of the chemical container **11** by the transfer member **111**, only one of the fall prevention guide or the alignment member **113** may be installed. For example, when the transport distance of the chemical container **11** is in a range of about 3000 mm to about 4000 mm, the chemical container **11** may be stably transferred by the guide, without the alignment member **113**.

(25) The inspection member **112** may be configured to inspect chemical information of the chemical container **11** from the other side of the transfer member **111**. The transfer member **111** may stop the operation when the chemical container **11** arrives at a correct position for chemical inspection. A sensor for confirming the position of the chemical container **11** may be installed on the transfer member **111** or the inspection member **112**. For example, the chemical carrying-in device **110** may include a second sensor **112a** for detecting whether the chemical container **11** or the pallet **12** protrudes from the other side of the transfer member **111**. The second sensor may be a fork sensor for detecting whether the chemical container **11** protrudes so that the chemical container **11** does not deviate from a position of the chemical inspection.

(26) The inspection member **112** may include an inspection sensor for inspecting chemical information of the chemical container **11**. For example, the inspection sensor may scan a QR code or barcode information attached to an upper portion or a side surface of the chemical container **11** to check material information, an expiration date, and the like, of the chemical container, and transmit the same to a control system. In addition, when the chemical information transmitted to the control system does not meet required specifications, the chemical container **11** may be returned externally by the transfer member **111**. The inspection member may include a housing or a frame structure through which the chemical container **11** is positioned, and which supports the inspection sensor. The inspection member may be described as an inspection compartment or an inspection area.

(27) An alignment member **113** may be installed on both sides of the transfer member **111** passing through the inspection member **112**. The alignment member **113** may align the chemical container **11** to a position for inspecting chemical information and carrying in and carrying out the chemical container **11** of the first robot (**140** in FIG. 3A). The alignment member **113** may include alignment bars or alignment arms that contact the pallet **12**, for example, and may be described as an alignment track.

(28) The chemical container **11** for which the chemical information is confirmed may be carried out from the chemical carrying-in device **110** by the first robot **140**. The chemical carrying-in device **110** may include an air spraying module for removing foreign substances before the chemical container **11** is carried out.

(29) FIGS. 3A and 3B are diagrams exemplarily illustrating a first robot **140** applicable to a chemical supply system of an example embodiment. FIG. 3A illustrates a state in which the first robot **140** carries out a chemical container bundle **10** from the chemical carrying-in device. FIG. 3B illustrates a state in which the chemical container bundle **10** is supported by a forklift **141** of the first robot **140**.

(30) Referring to FIGS. 3A and 3B, the first robot **140** according to an example embodiment may be configured to carry out the chemical container **11** carried into the chemical carrying-in device **110** and delivers the same to the temporary storage **120** or the second robot **150**. As an example, the first robot **140** may be a mobile robot **142** having a forklift **141**. The first robot **140** may raise and lower the chemical container bundle **10** by inserting the forklift **141** into the fork insertion portion of the pallet **12**. The forklift may have a two-pronged fork insertion portion, or may have forks with different numbers of prongs (e.g., 1, 3, or other amounts).

(31) The forklift **141** may have a certain level of effective payload (e.g., weight capacity) in order

to support the pallet **12** on which a plurality of chemical containers **11** are loaded. For example, the forklift **141** may have an effective payload of about 1200 kg or more.

(32) The mobile robot **142** may move to an operation position according to location information and/or path information input in advance according to a command of the control system. The mobile robot **142** may autonomously check a remaining battery level and move to a charging location to charge the battery.

(33) The first robot **140** may include a third sensor (refer to FIG. **4B**) for detecting a position of the pallet **12**. The third sensor may be installed on a side of the forklift **141**, and may be, for example, a vision sensor, a laser sensor, or a QR reader for detecting a position of the fork insertion portion of the pallet **12**, detecting horizontal and vertical alignment state of the pallet **12**, and the like. This will be described later with reference to FIGS. **4A** and **4B**.

(34) The chemical carrying-in device **110** may include a marker for inducing an approach of the first robot **140**. For example, the first robot **140** may detect a reflector, a QR code, or the like disposed on the other side of the transfer member **111** to approach the chemical carrying-in device **110**, and may insert the forklift **141** into the pallet **12** of the chemical container **11** loaded on the chemical carrying-in device **110**. Conversely, the chemical container **11** in which all of the chemicals are exhausted may be carried into the chemical carrying-in device **110** by the first robot **140** and carried out externally.

(35) In addition, the first robot **140** may include an information confirmation sensor (not shown) capable of checking raw material receipt and payment information by scanning a barcode, a QR code, and the like, attached to the chemical container **11** or the pallet **12**. The information confirmation sensor may be a laser sensor or a QR reader, installed on a side of the mobile robot **142** to confirm receipt and payment information of the chemical container.

(36) According to an example embodiment, the first robot **140** may directly insert the chemical container **11** into a chemical supply device **130**. In this case, the chemical supply device ('**130**' in FIGS. **6A** and **6B**) may include a conveyor line (not shown) for a buffer on which the chemical container **11** is seated. The chemical container **11** may be seated on one side of the buffer conveyor line in the chemical supply device ('**130**' in FIGS. **6A** and **6B**), and may be moved to an accommodating space of the chemical container by the buffer conveyor line. The first robot **140** may include control software and hardware that automatically controls the first robot **140** in response to control instructions received from a control computer or input by an operator. The software and hardware is configured to control all mechanical and other aspects of the first robot **140**, such as movement of wheels to move the first robot **140** to and from different locations, operation of the forklift, for example, to raise and lower the forklift, sensing operations, and other operations.

(37) FIGS. **4A** and **4B** are diagrams exemplarily illustrating a temporary storage **120** applicable to the chemical supply system of an example embodiment. FIGS. **4A** and **4B** illustrate a state in which the first robot **140** loads the chemical container bundle **10** on the temporary storage **120**.

(38) Referring to FIGS. **4A** and **4B**, the temporary storage **120** according to an example embodiment may have a chemical container storage space **121** in which the chemical container bundle **10** is accommodated. The temporary storage **120** provides a place for temporarily storing the chemical container bundle **10** being carried into the chemical carrying-in device and waiting for replacement, and the chemical container bundle **10** waiting to be carried out after all the chemicals are exhausted from the chemical supply device. The temporary storage **120** may be a two-stage or more pallet rack structure, for example, having a number of shelves and including a number of compartments, one for each container bundle **10**. The temporary storage **120** may be made of a corrosion-resistant coating or a corrosion-resistant material such as PTFE.

(39) The temporary storage **120** may include a position marker **M2** indicating a position of the chemical container storage space **121**. For example, the chemical container bundle **10** has a first position marker **M1** indicating a position of the pallet **12** or fork insertion portion, and the

temporary storage **120** may include a second position marker **M2** indicating a position of the chemical container storage space **121**. The first position marker **M1** and the second position marker **M2** may include a reflector, a QR code, a barcode, and the like. The position markers described herein may also be described as location markers.

(40) The first robot **140** may include a third sensor for recognizing the first position marker **M1** and the second position marker **M2**. The third sensor may be installed at an end **141a** of the forklift **141**. The third sensor may be a vision sensor, a laser sensor, or a QR reader, for example.

(41) According to an example embodiment, the temporary storage **120** may have an step or bump surrounding the chemical container storage space **121** so that the chemical container bundle **10** is not separated therefrom during a loading process of the chemical container bundle **10**. The step may be formed on rear and left and right side surfaces of the chemical container storage space **121** except for a front surface into which the chemical container bundle **10** is put.

(42) According to an example embodiment, the temporary storage **120** may include a fall prevention actuator (not shown) for supporting and fixing the chemical container bundle **10**. The fall prevention actuator may press the chemical container **11** or the pallet **12** from a side surface direction, block the front surface of the chemical container storage space **121**, or dock the chemical container **11** or the pallet **12** to fix the chemical container bundle **10** within a temporary storage **120** compartment.

(43) According to an example embodiment, the temporary storage **120** may include a load detection sensor for detecting whether the chemical container bundle **10** is transferred or loaded in the chemical container storage space **121**. The load detection sensor may transmit transfer and load information of the chemical container bundle **10** to the control system. For example, when the transfer and load information registered in the control system and the transfer information transmitted by the load detection sensor are different, it is possible to correct an error in the transfer and load information of the control system or to check whether the chemical container bundle **10** is deviated therefrom. In addition, the load detection sensor may be directly linked with the first robot **140** to prevent the first robot **140** from overloading the chemical container bundle **10** in the chemical container storage space **121**.

(44) According to an example embodiment, the temporary storage **120** may include a leakage detection sensor. The leakage detection sensor may detect whether a chemical leakage has occurred in the chemical container **11** stored in the temporary storage **120**.

(45) FIGS. 5A and 5B are diagrams exemplarily illustrating a second robot **150** applicable to a chemical supply system according to an example embodiment.

(46) Referring to FIGS. 5A and 5B, a second robot **150** of an example embodiment may be configured to put the chemical container **11** received from the first robot **140** into the chemical supply device ('**130**' in FIG. 6A). For example, the second robot **150** may be a mobile robot **152** having a conveyor **151** capable of transferring the chemical container **11** in one direction.

(47) The conveyor **151** may be configured as a roller or a belt. For example, the conveyor **151** may transport the chemical container **11** in one direction or in a direction opposite to the one direction by rotating a plurality of rollers disposed in one direction.

(48) Depending on example embodiments, a load detection sensor for detecting whether the chemical container **11** or the pallet **12** is seated on the conveyor **151** may be disposed. For example, in order to determine whether the chemical container **11** is seated and the position of the conveyor **151**, a load detection sensor may be installed at regular intervals on a side of the conveyor **151** or the fall prevention guide. In addition, the conveyor **151** may be made of a corrosion-resistant coating or a corrosion-resistant material such as PTFE, and according to example embodiments, a leak detection sensor may be installed on the conveyor **151** to detect any chemical leakage of the chemical container **11**.

(49) The second robot **150** may further include an alignment member **153** configured to align the chemical container bundle **10** on the conveyor **151**. The alignment member **153** may be disposed

on both sides of the conveyor **151** and in a direction of movement of the pallet **12**, and may align the chemical container **11** or the pallet **12** to a central portion of the conveyor **151**. For example, the alignment member **153** may align the chemical container **11** to the central portion of the conveyor **151** before the conveyor **151** operates. The alignment member **153** may have a similar structure to the alignment member **113** discussed above in connection with transfer member **111**. In addition, guides for preventing separation of the chemical container **11** or the chemical container bundle **10** may be installed on both sides of the conveyor **151** for use while the conveyor **151** is operating.

(50) The mobile robot **152** may move to an operating position according to position information and/or path information input in advance according to a command of a control system. The mobile robot **152** may autonomously determine a remaining battery level and move to a charging location to charge the battery.

(51) The second robot **150** may include a fourth sensor **152a** for recognizing a chemical container accommodating space of the chemical supply device ('**130**' in FIG. **6A**). The fourth sensor may be installed on a side of the mobile robot **152** to detect a symbol, reflector or code such as a QR code attached to an outside or inside of the chemical supply device ('**130**' in FIG. **6A**). The fourth sensor **152a** may be a vision sensor, a laser sensor, or a QR reader, for example. When the first robot **140** puts the chemical container into the chemical supply device ('**130**' in FIG. **6A**) using a forklift **141**, it may be necessary to install a conveyor line for a buffer inside the chemical supply device ('**130**' in FIG. **6A**) to prevent damage to internal devices and chemical containers. However, since the second robot **150** directly puts the chemical container **11** in the chemical supply device using the conveyor **151**, a conveyor line for the buffer is not needed, and it is possible to reduce manufacturing costs of the chemical supplying device ('**130**' in FIG. **6A**) and to dispose a larger number of chemical supply devices ('**130**' in FIG. **6A**).

(52) FIGS. **6A** and **6B** are diagrams exemplarily illustrating a chemical supply device **130** applicable to a chemical supply system according to an example embodiment.

(53) Referring to FIGS. **6A** and **6B**, the chemical supply device **130** of an example embodiment may connect a chemical supply pipe **22** to an outlet of the chemical container **11** accommodated in the chemical container accommodating space **131**, to supply chemicals in the chemical container **11** to various semiconductor processing operations. A loader unit, an alignment unit, and the like, capable of supporting the chemical container **11** and adjusting a position thereof may be disposed in the chemical container accommodating space **131**. As illustrated in the figure, a pallet **12** and chemical containers **11** may be carried into the chemical supply device **130** together, but depending on example embodiments, only the chemical containers **11** without the pallet **12** may be carried thereinto.

(54) The chemical supply device **130** may include a coupler connection module (not illustrated) for automatically fastening and separating the coupler **21** to an outlet of the chemical container **11**. One end of the coupler **21** may be connected to the chemical supply pipe **22** for distributing the chemical. The coupler connection module may include an opening/closing unit for opening and closing a cap of the chemical container, a coupler unit for fastening and separating the coupler **21** to the outlet of the chemical container, a control unit for controlling an operation thereof, and the like. Depending on example embodiments, the coupler **21** may be automatically coupled to and separated from the chemical container **11** by a mobile robot having a robot arm (refer to an example embodiment of FIG. **7A**).

(55) In addition, a chemical container **11** in which all chemicals are exhausted may be replaced with a new chemical container **11** by a second robot **150**. The second robot **150** may perform an operation according to a transfer command, a replacement command, and the like of the chemical container transmitted from a control system. The second robot **150** may access the chemical supply device **130** designated by the control system, and perform a transfer operation of the chemical container bundle **10**. The second robot **150** may perform a transfer operation of the chemical container bundle **10** according to position information and/or path information input in advance. A

reflector or QR code including the position information thereof may be attached to an outside or an inside of the chemical supply device **130**. The second robot **150** may identify the chemical supply device **130** by using a vision sensor, a laser sensor, or a QR reader, and perform carrying-in and carrying-out operations of the chemical container **11** from the chemical supply device **130** designated by the control system. According to example embodiment, the chemical supply device may include a communication unit transmitting carrying-in and carrying-out information of the chemical container to the second robot **150**. The accommodating space **131** may include a shelf or platform, generally described as a floor or bottom surface, on which the pallet **12** is placed by the second robot **150**.

(56) FIGS. 7A to 7D are diagrams exemplarily illustrating a third robot **160** applicable to a chemical supply system according to an example embodiment.

(57) Referring to FIGS. 7A to 7D, a chemical supply device **130a** of an example embodiment may include a coupler module **135** and a cradle **132**, and the chemical container **11** carried into the chemical accommodating space **131a** may be coupled to and separated from the coupler module **135** by the third robot **160**. The chemical containers **11** individually carried in the chemical supply device **130a** may be fixed to a support member **133** disposed on one side of the third robot **160**. The support member **133** may support and fix the chemical container **11** in a process of coupling and separating the coupler module **135**. However, depending on an example embodiment, the pallet **12** and the chemical containers **11** may be carried thereinto together, and so either no support member **133** is included, or the support member **133** may be pre-installed on a pallet **12**.

Alternatively, a fourth robot may be used to individually separate the chemical containers **11** from the pallet **12** and place them in the chemical accommodating space **131a** of FIGS. 7A-7D, or the third robot **160** and a coupler module **135** and cradle **132** such as depicted in FIGS. 7A-7D may be used in a chemical supply device such as chemical supply device **130** of FIGS. 6A-6B. The third robot **160** may perform an operation according to a replacement command of the chemical container transmitted from a control system. The third robot **160** may access the chemical supply device **130a** designated by the control system to perform a replacement operation of the chemical container **11**. The control system may optimize a movement path and operating performance of the third robots **160**, by using the position information and operating information of the plurality of third robots **160**.

(58) The chemical supply device **130a** and the third robot **160** of the present example embodiment do not necessarily constitute a chemical supply system together with the equipment shown in FIGS. 2A to 6B. For example, the chemical supply system of an example embodiment may include a chemical supply device **130a** having a chemical container accommodating space **131a** in which a chemical container **11** in which chemical information is confirmed is accommodated, and configured to supply the chemical in the chemical container **11** to a place of use thereof; and a mobile robot **162** configured to fasten and separate the coupler module **135** to an outlet of the chemical container **11** carried in the chemical accommodating space **131a**.

(59) The cradle **132** may be disposed on one side of the chemical container accommodating space **131a**. The cradle **132** may store the coupler module **135** that is not fastened to the chemical container **11**. The cradle **132** may be connected to a chemical drain **132a** for removing chemicals remaining in the coupler module **135**. In addition, a cap holder **132b** capable of temporarily holding the cap **11a** of the chemical container **11** to which the coupler module **135** is fastened may be installed on one side of the cradle **132**.

(60) The coupler module **135**, also described as a coupling interface, may include a main housing, a grip portion **135a**, also described as a gripper receiving slot, gripped by the gripper **161a**, also described as a gripping claw, a driving force transmission unit **135b**, also described as a driving transmission, provided with driving force from the robot arm **161**, and a coupler driving unit **135c** receiving driving force from the driving force transmission unit **135b** to fasten and separate the coupler **21** (see FIG. 7C). For example, the driving force transmission unit **135b** may be a gear unit

docked with the gripper **161a** or a power transmission link inside the gripper **161a**, to transmit rotational force. The coupler module **135** may also include a coupler connection portion where a coupler **21** connects to the coupler module **135**.

(61) One end of the coupler **21** may be connected to the chemical supply pipe **22** for distributing chemicals. A key-code may be formed in a fastening portion of the coupler **21**. The coupler **21** may be fastened to an outlet of the chemical container **11** in which the key-code corresponding thereto is formed. The coupler **21** may be, for example, a coupling nozzle and/or ring or a coupling terminal having a circular or other shape that fits at one end around the chemical supply pipe **22** and outputs the chemical at the other end.

(62) For example, the coupler **21** may have an embossed key-code, and the chemical container **11** may have an engraved key-code. Since the key-code of the chemical container **11** is formed differently depending on the type of chemical, the coupler **21** and the chemical container **11** of which key code does not correspond to each other may not be fastened to each other.

(63) The third robot **160** may be a mobile robot **162** having a robot arm **161** configured to grip the coupler module **135**. The third robot **160** may be made of a corrosion-resistant coating or a corrosion-resistant material such as PTFE. According to an embodiment, an air spraying module may be installed to prevent the third robot **160** from being exposed to chemicals.

(64) The robot arm **161** may have a gripper **161a** for gripping the coupler module **135** at one end, and may be configured to provide driving force for fastening and separating the coupler module **135**. The robot arm **161** may have an articulated arm shape to perform a coupling operation of the coupler module **135** in the chemical container accommodating space **131**. A vision sensor, and the like may be installed together with the gripper **161a** at one end of the articulated arm.

(65) The mobile robot **162** may move to an operating position according to position information and/or path information input in advance according to a command of the control system. The mobile robot **162** may sense a reflector or QR code attached to the chemical supply device **130a** to access the chemical supply device **130a** used for an operation. The mobile robot **162** may autonomously check the remaining battery level and move to a charging location to charge the battery. A labyrinth seal structure blocking chemical penetration may be applied to a charging terminal of a charging station.

(66) The third robot **160** may be configured to perform operations such as monitor operation, door opening and closing, cap opening and closing, coupler module coupling, chemical leakage cleaning, and the like.

(67) For example, when the third robot **160** approaches a chemical supply device **130a** requiring an operation, the robot arm **161** may sense a monitor of the chemical supply device **130a** using a vision sensor, operate the monitor with the gripper **161a**, and change the same to an operating mode of the chemical supply device **130a**.

(68) As another example, when the third robot **160** is switched to a replacement mode of the chemical container **11**, in which a replacement chemical container **11** is transferred into a chemical supply device **130a**, the third robot **160** may open a door of the chemical supply device **130a** using the gripper **161a**. However, a replacement operation of the chemical container **11** using the third robot **160** does not need to use a vision sensor. According to an example embodiment, the replacement operation of the chemical container **11** may be performed by recognizing a QR code attached to the cradle **132**, the coupler module **135**, the door of the chemical supply device **130a**, the chemical container **11**, and the like.

(69) As another example, the robot arm **161** may sense a cap **11a** and a drum outlet of the chemical container **11** with a vision sensor, to check position information of the cap **11a**. The robot arm **161** may open and close the cap by using the position information of the cap **11a**.

(70) For example, the robot arm **161** may be configured to rotate the gripper **161a** or a power transmission link inside the gripper **161a**. The robot arm **161** may grip the cap **11a** using the gripper **161a**, and transmit a rotational force to the cap **11a** to separate the cap **11a** from the

chemical container **11**.

(71) In one embodiment, a compliance joint may be installed on one side of the gripper **161a** so that the cap **11a** can be fastened and separated even when an upper surface of the chemical container **11** is slightly inclined (for example, about $\pm 5^\circ$) with respect to a horizontal plane.

(72) The robot arm **161** may sense a cradle **132** with a vision sensor, and mount the cap **11a** on a cap holder **132b**.

(73) In one embodiment, the robot arm **161** may grip the coupler module **135** stored in the cradle **132**, remove it from the cradle **132**, and fasten the coupler module **135** to an outlet of the chemical container **11** from which the cap **11a** is carried out. The robot arm **161** may fasten the coupler module **135** by using position information of the outlet of the chemical container **11** confirmed by the vision sensor. The robot arm **161** may confirm a key code of each of the coupler **21** and the chemical container **11** with a vision sensor, align the key code, or confirm whether the key codes correspond to each other.

(74) In one embodiment, the robot arm **161** may detect a chemical leakage generated during replacement of the chemical container **11** and wipe it. The robot arm **161** may grip or dock a cleaning material provided in the chemical supply device **130a** to wipe the chemical leakage. The robot arm **161** may perform the above-described operations in a reverse order to separate the coupler module **135** from the chemical container **11** in which all the chemicals are exhausted, and fasten the cap **11a**. During an entire process, it is possible to determine whether the coupler module **135** and the cap **11a** are properly fastened, or fastened with an appropriate torque using a laser sensor, a torque sensor, and the like included in the robot arm **161**.

(75) The robot arm **161** may perform an operation such as shaking or turning over the separated coupler module **135** in order to minimize chemical leakage. In addition, the robot arm **161** may store the separated coupler module **135** in the cradle **132**.

(76) FIG. **8** is a diagram illustrating a chemical supply system **100A** comprised of the exemplary devices shown in FIGS. **2A** to **6B**.

(77) Referring to FIG. **8**, a chemical supply system **100A** may perform a replacement operation of a chemical container using a first robot **140** of FIGS. **3A** and **3B** and a second robot **150** of FIGS. **5A** and **5B**. The first robot **140** and the second robot **150** may perform an operation in each operation according to an operation command transmitted from a control system, thereby preventing safety accidents and improving efficiency of a transfer operation and a replacement operation of the chemical container. The control system can include one or more of the following components: at least one central processing unit (CPU) configured to execute computer program instructions to perform various processes and methods, random access memory (RAM) and read only memory (ROM) configured to access and store data and information and computer program instructions, input/output (I/O) devices configured to provide input and/or output to the processing controller **1020** (e.g., keyboard, mouse, display, speakers, printers, modems, network cards, etc.), and storage media or other suitable type of memory (e.g., such as, for example, RAM, ROM, programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic disks, optical disks, floppy disks, hard disks, removable cartridges, flash drives, any type of tangible and non-transitory storage medium) where data and/or instructions can be stored. In addition, control system can include antennas, network interfaces that provide wireless and/or wire line digital and/or analog interface between components and to one or more networks over one or more network connections (not shown), a power source that provides an appropriate alternating current (AC) or direct current (DC) to power one or more components, and a bus that allows communication among the various disclosed components.

(78) In a first operation (S1), a chemical container bundle **10** may be carried into a chemical carrying-in device **110**. The chemical carrying-in device **110**, also described as a chemical loading device, may inspect chemical information of the chemical containers **11**, and transmit the same to

the control system. For example, when the chemical information is inappropriate, the chemical carrying-in device **110** may be operated to carry out the chemical container bundle **10** externally. The chemical container bundle **10** for which the chemical information is confirmed may be carried out from the chemical carrying-in device **110** by the first robot **140**. The first robot **140** may transfer the chemical container bundle **10** from the chemical carrying-in device **110** to a temporary storage **120** or transfer the same to the second robot **150**.

(79) In a second operation (S2), the temporary storage **120** may store chemical container bundles **10** carried into the chemical carrying-in device **110** and waiting to replace existing chemical container bundles **10** when they are empty, as well as chemical container bundles **10** that have been emptied of all the chemicals and waiting to be carried out. The first robot **140** may deliver a replacement chemical container bundle **10** from among the chemical container bundles **10** stored in the temporary storage **120** to the second robot **150**. Alternatively, the first robot **140** may return a chemical container bundle **10** in which the chemical is exhausted (e.g., emptied) to the chemical carrying-in device **110**, and the chemical container bundle **10** returned by the first robot **140** may be carried out externally by the chemical carrying-in device **110**.

(80) In a third operation (S3), the second robot **150** may be configured to carry in a chemical container bundle **10** received from the first robot **140** into the chemical supply device **130**. The second robot **150** may transfer the chemical container bundle **10** or a chemical container received from the first robot **140** to the chemical supply device **130**. The chemical supply device **130** may connect a chemical supply pipe to an outlet of the chemical container **11** carried thereinto, to supply the chemicals in the chemical container **11** to various semiconductor process operations.

(81) The mobile robots described herein include wheels or other transportation means for moving freely about a facility and include a control system configured to provide autonomous movement based on received commands or instructions.

(82) As set forth above, according to example embodiments of the present inventive concept, by automating carrying-in and carrying-out operations of a chemical container, a chemical supply system having improved operational efficiency as well as improved safety may be provided.

(83) While example embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present invention as defined by the appended claims.

Claims

1. A chemical supply system, comprising: a chemical container receiving station configured to receive a chemical container, inspect chemical information of the chemical container and confirm the chemical information; a temporary storage configured to store the chemical container; a chemical supply device configured to supply a chemical in the chemical container to a place of use of the chemical; a first mobile robot configured to receive the chemical container for which the chemical information is confirmed from the chemical container receiving station and to transport the chemical container freely and autonomously along a floor of a facility, the first mobile robot including a lift configured to raise and lower a pallet upon which the chemical container is loaded; and a second mobile robot configured to carry the chemical container transported by the first mobile robot into the chemical supply device, the second mobile robot including a conveyor configured to receive the pallet from the lift of the first mobile robot.

2. The chemical supply system of claim 1, wherein the chemical container receiving station comprises: a transfer member having a first side on which the chemical container is seated when received and a second side to which the chemical container is transferred in order to inspect the chemical information; and an inspection area for inspecting the chemical information on the second side.

3. The chemical supply system of claim 2, wherein, when the pallet on which the chemical

container is loaded is on the first side of the transfer member, the transfer member is operated so that the pallet is transferred to the second side, and when the pallet is seated on the second side of the transfer member, the transfer member is stopped.

4. The chemical supply system of claim 3, wherein the chemical container receiving station comprises a first sensor for detecting whether the pallet is seated on the first side of the transfer member.

5. The chemical supply system of claim 4, wherein the chemical container receiving station comprises a second sensor for detecting whether the pallet protrudes from the second side of the transfer member.

6. The chemical supply system of claim 2, wherein the chemical container receiving station further comprises an alignment member configured to align the pallet on which the chemical container is loaded on the transfer member.

7. The chemical supply system of claim 1, wherein the first mobile robot is configured to deliver the chemical container to the temporary storage in addition to being configured to deliver the chemical container to the second mobile robot.

8. The chemical supply system of claim 1, wherein the first mobile robot has a forklift.

9. The chemical supply system of claim 8, wherein: the pallet has a fork insertion portion, and the first mobile robot is configured to insert the forklift into the fork insertion portion to raise and lower the pallet.

10. The chemical supply system of claim 8, wherein the first mobile robot comprises a sensor for recognizing a storage space of the temporary storage, the storage space for storing the chemical container.

11. The chemical supply system of claim 1, wherein the second mobile robot is configured to put the chemical container received from the first mobile robot into the chemical supply device.

12. The chemical supply system of claim 1, wherein the second mobile robot comprises a sensor for recognizing an accommodating space of the chemical supply device for the chemical container.

13. The chemical supply system of claim 1, wherein: the chemical supply device comprises a coupler connection module for fastening a coupler to an outlet of the chemical container, and the coupler is connected to a chemical supply pipe for distributing the chemical in the chemical container.

14. The chemical supply system of claim 1, further comprising: a third mobile robot configured to fasten and separate a coupler module to an outlet of the chemical container transported into the chemical supply device.

15. The chemical supply system of claim 14, wherein the third mobile robot has a robot arm configured to grip the coupler module.

16. The chemical supply system of claim 14, wherein: the third mobile robot is configured to provide driving force for fastening and separating the coupler module to the outlet, and the coupler module comprises a coupler connected to a chemical supply pipe for distributing the chemical in the chemical container, and a driving force transmission unit configured to transmit the driving force to the coupler.

17. A chemical supply system, comprising: a chemical supply device having a chemical container accommodating space in which a chemical container is accommodated, and configured to supply a chemical in the chemical container to a place of use of the chemical; and a mobile robot configured to fasten and separate a coupling interface to an outlet of the chemical container carried into the chemical accommodating space, the mobile robot being configured to autonomously and freely move about a floor of a facility, wherein: the mobile robot has a robot arm configured to grip the coupling interface, and the coupling interface comprises a coupling terminal connected to a chemical supply pipe, and a driving force transmission unit configured to transmit driving force provided from the robot arm to the coupling terminal.

18. A chemical supply system, comprising: a chemical container receiving station configured to

inspect chemical information of a chemical container and confirm the chemical information; a temporary storage configured to store the chemical container; a chemical supply device configured to supply a chemical in the chemical container to a place of use of the chemical; first and second mobile robots configured to transfer the chemical container for which the chemical information is confirmed to the temporary storage or the chemical supply device, wherein the first mobile robot is configured to transfer the chemical container from the chemical container receiving station to the temporary storage and to transfer the chemical container to the second mobile robot, and wherein the second mobile robot is configured to receive the chemical container from the first mobile robot and transfer the chemical container received from the first mobile robot to the chemical supply device.
